

PAVAN KUMAR KETHA

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★ Visakhapatnam, Andhra Pradesh

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TECHNICAL SUMMARY

Computer Science student with hands-on experience developing and deploying practical technology solutions. Built projects across artificial intelligence, software, and IoT, applying machine learning, information retrieval, and backend development to solve real-world problems.

PROJECTS

Disaster Insight Hub Assistant

- Developed and deployed an AI-powered disaster guidance application that classifies crisis reports across **7 categories**, using a Random Forest model trained on **2,345 labelled records**.
- Implemented a retrieval-augmented guidance pipeline using **TF-IDF retrieval** over a curated knowledge base to generate contextually relevant disaster response information.
- Integrated **spaCy-based location extraction** and a cloud LLM through **REST APIs** to generate location-aware, context-driven responses within a Flask application.
- Containerized the application using **Docker** and deployed it as a publicly accessible web application.

Live Demo: <https://diha.vercel.app>

Tech Stack: Python, Flask, scikit-learn, TF-IDF, spaCy, NLP, RAG, REST APIs, Docker, LLM API Integration

Repository Link: <https://github.com/Pavan-5905/Disaster-Insight-Hub-Assistant>

Smart Weather Monitoring System

- Developed an offline IoT-based weather monitoring system using **ESP8266** and **DHT11** sensors to provide real-time temperature and humidity data without relying on internet connectivity.
- Implemented periodic sensor data collection with readings refreshed every **2** seconds and displayed the environmental information through an **LCD** interface.
- Designed the system for practical use in rural farming environments, enabling local access to environmental data through an embedded hardware setup.

Tech Stack: C++, ESP8266, DHT11, Internet of Things (IoT), Embedded Systems

Repository Link: <https://github.com/Pavan-5905/Smart-Weather-Monitoring-System>

TECHNICAL SKILLS

- **Programming:** Python, Java, SQL
- **Core Concepts:** DSA, OOP
- **Web Technologies:** HTML, CSS, JavaScript
- **AI Concepts:** RAG
- **Tools & Platforms:** GitHub, Docker, Vercel, Railway, Cursor

EDUCATION

B. TECH – Computer Science and Engineering
Dadi Institute of Engineering & Technology | Anakapalle

- 7th Semester
Percentage: 72.3%

CERTIFICATIONS

TCS ION NQT-IT – **72.53%**
Python – *Elite + Silver*
Front-End Web Development

TCS ION Jul, 2026
NPTEL Nov, 2025
eDX | IBM Jul, 2024

ACCOMPLISHMENTS

- **Student Chair, CSI Student Branch (DIET), 2025–2026:** Led the student branch, organizing 12 technical events including a national-level conference.
- **Git & GitHub Workshop Facilitator:** Conducted a hands-on workshop on Git fundamentals, repository management, and branching workflows, helping participants understand collaborative version control practices.
- **Security Coordination Team, CII Partnership Summit (Visakhapatnam, Nov 2025):** Volunteered in event security coordination for the summit.
- **Community Service:** Led an 8-week organic farming awareness camp (Apr–May 2025), teaching local farmers sustainable practices including bio-fertilizers, composting, and water harvesting.

COMMUNITIES

- **LinkedIn:** www.linkedin.com/in/pavan-kumar-ketha
- **GitHub:** github.com/Pavan-5905
- **HackerRank:** www.hackerrank.com/k_pavan_5905

Date: 24th August, 2026

Place: Visakhapatnam, Andhra Pradesh